

DERIVADAS

1) $y = cte$	$y' = 0$	Derivada de una función de función $y = f(u); u = f(x)$	
2) $y = x$	$y' = 1$	2) $y = u$	$y' = u'$
3) $y = c \cdot x$	$y' = c$	3) $y = c \cdot u$	$y' = c \cdot u'$
Derivada de una función $\lim_{x \rightarrow a} f(x) = f'(x) = \frac{dy}{dx}; a \in \mathbb{R}; y = f(x)$		4) $y = u \pm v \pm w$	$y' = u' \pm v' \pm w'$
		5) $y = u \cdot v$	$y' = u' \cdot v + u \cdot v'$
		6) $y = \frac{u}{v}$	$y' = \frac{u' \cdot v - u \cdot v'}{v^2}$
		7) $y = u^v$	$y' = u^v \left[v' \ln(u) + v \cdot \frac{u'}{u} \right]$
8) $y = x^m$	$y' = m \cdot x^{m-1}$	8) $y = u^m$	$y' = m \cdot u^{m-1} \cdot u'$
9) $y = \sqrt[m]{x}$	$y' = \frac{1}{m \sqrt[m]{x^{m-1}}}$	9) $y = \sqrt[m]{u}$	$y' = \frac{u'}{m \sqrt[m]{u^{m-1}}}$
10) $y = \sqrt{x}$	$y' = \frac{1}{2\sqrt{x}}$	10) $y = \sqrt{u}$	$y' = \frac{u'}{2\sqrt{u}}$
11) $y = \log_a(x)$	$y' = \frac{1}{x} \log_a e$	11) $y = \log_a(u)$	$y' = \frac{u'}{u} \log_a e$
12) $y = \ln(x)$	$y' = \frac{1}{x}$	12) $y = \ln(u)$	$y' = \frac{u'}{u}$
13) $y = a^x$	$y' = a^x \ln(a)$	13) $y = a^u$	$y' = a^u \cdot u' \cdot \ln(a)$
14) $y = e^x$	$y' = e^x$	14) $y = e^u$	$y' = u' \cdot e^u$
15) $y = \text{sen}(x)$	$y' = \cos(x)$	15) $y = \text{sen}(u)$	$y' = \cos(u) \cdot u'$
16) $y = \cos(x)$	$y' = -\text{sen}(x)$	16) $y = \cos(u)$	$y' = -\text{sen}(u) \cdot u'$
17) $y = \text{tg}(x)$	$y' = \frac{1}{\cos^2(x)} = 1 + \text{tg}^2(x)$ $= \sec^2(x)$	17) $y = \text{tg}(u)$	$y' = \frac{u'}{\cos^2(u)} = u'(1 + \text{tg}^2(u))$ $= \sec^2(u) \cdot u'$
18) $y = \text{cotg}(x)$	$y' = \frac{-1}{\text{sen}^2(x)} = -\text{cosec}^2(x)$ $= -(1 + \text{cotg}^2(x))$	18) $y = \text{cotg}(u)$	$y' = \frac{-u'}{\text{sen}^2(u)} = -u'(1 + \text{cotg}^2(u))$ $= -\text{cosec}^2(u) \cdot u'$
19) $y = \sec(x)$	$y' = \sec(x) \cdot \text{tg}(x)$	19) $y = \sec(u)$	$y' = \sec(u) \cdot \text{tg}(u) \cdot u'$
20) $y = \text{cosec}(x)$	$y' = -\text{cosec}(x) \cdot \text{cotg}(x)$	20) $y = \text{cosec}(u)$	$y' = -\text{cosec}(u) \cdot \text{cotg}(u) \cdot u'$
21) $y = \arcsen(x)$	$y' = \frac{1}{\sqrt{1-x^2}}$	21) $y = \arcsen(u)$	$y' = \frac{u'}{\sqrt{1-u^2}}$
22) $y = \arccos(x)$	$y' = \frac{-1}{\sqrt{1-x^2}}$	22) $y = \arccos(u)$	$y' = \frac{-u'}{\sqrt{1-u^2}}$
23) $y = \text{arctg}(x)$	$y' = \frac{1}{1+x^2}$	23) $y = \text{arctg}(u)$	$y' = \frac{u'}{1+u^2}$
24) $y = \text{arccotg}(x)$	$y' = \frac{-1}{1+x^2}$	24) $y = \text{arccotg}(u)$	$y' = \frac{-u'}{1+u^2}$
25) $y = \text{arcsec}(x)$	$y' = \frac{1}{x\sqrt{x^2-1}}$	25) $y = \text{arcsec}(u)$	$y' = \frac{u'}{u\sqrt{u^2-1}}$
26) $y = \text{arccosec}(x)$	$y' = \frac{-1}{x\sqrt{x^2-1}}$	26) $y = \text{arccosec}(u)$	$y' = \frac{-u'}{u\sqrt{u^2-1}}$

INTEGRALES

Integrales Inmediatas de una Variable

$$1) \int (f_1(x) \pm f_2(x) \pm \dots \pm f_n(x)) dx = \int f_1(x) dx \pm \int f_2(x) dx \pm \dots \pm \int f_n(x) dx$$

$$3) \int dx = x + c$$

$$4) \int x^m \cdot dx = \frac{x^{m+1}}{m+1} + c$$

$$5) \int \frac{1}{x} \cdot dx = \ln|x| + c$$

$$6) \int e^x \cdot dx = e^x + c$$

$$7) \int a^x \cdot dx = \frac{a^x}{\ln(a)} + c$$

$$8) \int \sin(x) \cdot dx = -\cos(x) + c$$

$$9) \int \cos(x) \cdot dx = \sin(x) + c$$

$$10) \int \tan(x) \cdot dx = \ln|\sec(x)| + c = -\ln|\cos(x)| + c$$

$$11) \int \operatorname{cosec}(x) \cdot dx = \ln|\operatorname{cosec}(x) - \cotg(x)| + c$$

$$12) \int \sec(x) \cdot dx = \ln|\sec(x) + \tan(x)| + c$$

$$13) \int \cotg(x) \cdot dx = \ln|\sin(x)| + c$$

$$14) \int \frac{1}{\cos^2(x)} \cdot dx = \int \sec^2(x) \cdot dx = \tan(x) + c$$

$$15) \int \frac{1}{\sin^2(x)} \cdot dx = \int \operatorname{cosec}^2(x) \cdot dx = -\cotg(x) + c$$

$$16) \int \sec(x) \cdot \tan(x) dx = \sec(x) + c$$

$$17) \int \operatorname{cosec}(x) \cdot \cotg(x) dx = -\operatorname{cosec}(x) + c$$

$$18) \int \frac{1}{\sqrt{1-x^2}} \cdot dx = \begin{cases} \arcsen(x) + c \\ -\arccos(x) + c \end{cases}$$

$$19) \int \frac{1}{1+x^2} \cdot dx = \begin{cases} \arctg(x) + c \\ -\operatorname{arccotg}(x) + c \end{cases}$$

$$20) \int \frac{1}{x\sqrt{x^2-1}} \cdot dx = \begin{cases} \operatorname{arcsec}(x) + c \\ -\operatorname{arccosec}(x) + c \end{cases}$$

Integrales Inmediatas de Funciones

$$u = f(x)$$

$$2) \int u \cdot dv = u \cdot v - \int v \cdot du \quad \text{Integración por partes}$$

$$4) \int u^m \cdot u' \cdot dx = \frac{u^{m+1}}{m+1} + c$$

$$5) \int \frac{u'}{u} \cdot dx = \ln|u| + c$$

$$6) \int e^u \cdot u' \cdot dx = e^u + c$$

$$7) \int a^u \cdot u' \cdot dx = \frac{a^u}{\ln(a)} + c$$

$$8) \int \sin(u) \cdot u' \cdot dx = -\cos(u) + c$$

$$9) \int \cos(u) \cdot u' \cdot dx = \sin(u) + c$$

$$10) \int \tan(u) \cdot u' \cdot dx = \ln|\sec(u)| + c = -\ln|\cos(u)| + c$$

$$11) \int \operatorname{cosec}(u) \cdot u' \cdot dx = \ln|\operatorname{cosec}(u) - \cotg(u)| + c$$

$$12) \int \sec(u) \cdot u' \cdot dx = \ln|\sec(u) + \tan(u)| + c$$

$$13) \int \cotg(u) \cdot u' \cdot dx = \ln|\sin(u)| + c$$

$$14) \int \frac{u'}{\cos^2(u)} \cdot dx = \int \sec^2(u) \cdot dx = \tan(u) + c$$

$$15) \int \frac{u'}{\sin^2(u)} \cdot dx = \int \operatorname{cosec}^2(u) \cdot dx = -\cotg(u) + c$$

$$16) \int \sec(u) \cdot \tan(u) \cdot u' \cdot dx = \sec(u) + c$$

$$17) \int \operatorname{cosec}(u) \cdot \cotg(u) \cdot u' \cdot dx = -\operatorname{cosec}(u) + c$$

$$18) \int \frac{1}{\sqrt{1-u^2}} \cdot u' \cdot dx = \begin{cases} \arcsen(u) + c \\ -\arccos(u) + c \end{cases}$$

$$19) \int \frac{1}{1+u^2} \cdot u' \cdot dx = \begin{cases} \arctg(u) + c \\ -\operatorname{arccotg}(u) + c \end{cases}$$

$$20) \int \frac{1}{u\sqrt{u^2-1}} \cdot u' \cdot dx = \begin{cases} \operatorname{arcsec}(u) + c \\ -\operatorname{arccosec}(u) + c \end{cases}$$

RELACIONES TRIGONOMÉTRICAS

Suma y Diferencia de Ángulos

$$\sin(\alpha + \beta) = \sin\alpha \cdot \cos\beta + \cos\alpha \cdot \sin\beta$$

$$\sin(\alpha - \beta) = \sin\alpha \cdot \cos\beta - \cos\alpha \cdot \sin\beta$$

$$\cos(\alpha + \beta) = \cos\alpha \cdot \cos\beta - \sin\alpha \cdot \sin\beta$$

$$\cos(\alpha - \beta) = \cos\alpha \cdot \cos\beta + \sin\alpha \cdot \sin\beta$$

$$\operatorname{tg}(\alpha + \beta) = \frac{\operatorname{tg}\alpha + \operatorname{tg}\beta}{1 - \operatorname{tg}\alpha \cdot \operatorname{tg}\beta}$$

$$\operatorname{tg}(\alpha - \beta) = \frac{\operatorname{tg}\alpha - \operatorname{tg}\beta}{1 + \operatorname{tg}\alpha \cdot \operatorname{tg}\beta}$$

Identidades de Ángulos

$$\sin(2\alpha) = 2 \cdot \sin\alpha \cdot \cos\alpha$$

$$\cos(2\alpha) = \cos^2\alpha - \sin^2\alpha$$

$$\operatorname{tg}(2\alpha) = \frac{2 \cdot \operatorname{tg}\alpha}{1 - \operatorname{tg}^2\alpha}$$

Identidad Pitagórica

$$\sin^2\alpha + \cos^2\alpha = 1$$

Suma a Producto

$$\sin\alpha \pm \sin\beta = 2 \sin\frac{1}{2}(\alpha \pm \beta) \cos\frac{1}{2}(\alpha \mp \beta)$$

$$\cos\alpha + \cos\beta = 2 \cos\frac{1}{2}(\alpha + \beta) \cos\frac{1}{2}(\alpha - \beta)$$

$$\cos\alpha - \cos\beta = -2 \sin\frac{1}{2}(\alpha + \beta) \cos\frac{1}{2}(\alpha - \beta)$$